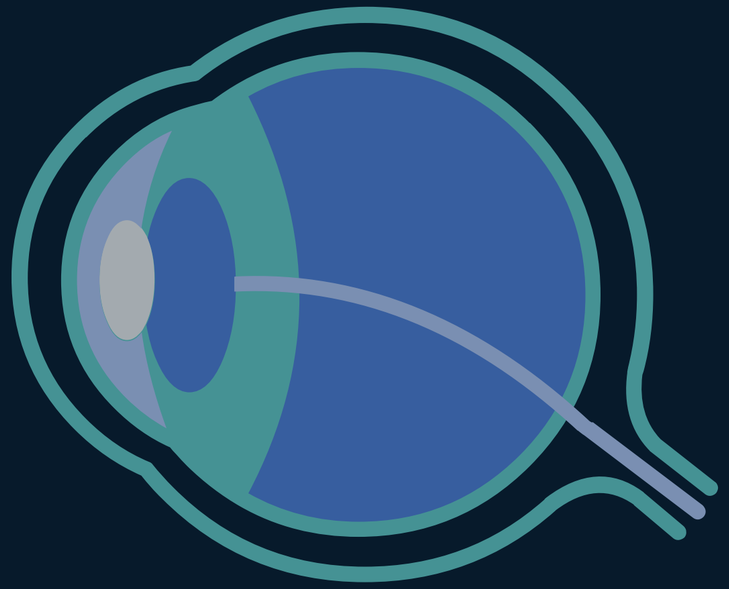
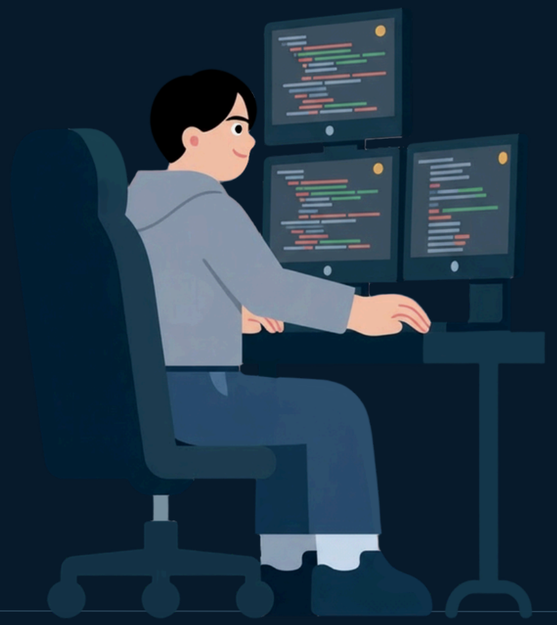
CSE427 PROJECT PRESENTATION

AUTOMATED DIABETIC RETINOPATHY SEVERITY CLASSIFICATION

USING RETINAL FUNDUS IMAGES

RESNET-50 ARCHITECTURE





Pretom Areefin Pranto • MD. Farhad Hossain Shishir

TWO-MEMBER WORK DISTRIBUTION

JOINT

BALANCED OWNERSHIP ACROSS DATA, MODELING, EVALUATION AND INTERPRETATION.

PRETOM AREEFIN PRANTO

DATA PIPELINE + MODEL SETUP

- DATASET LOADING, IMAGE-PATH MATCHING AND DATA-QUALITY CHECKS
- SEVERE-CLASS REMOVAL AND PDR LABEL REMAPPING
- EDA, PREPROCESSING AND TRAIN-ONLY AUGMENTATION
- STRATIFIED 80/10/10 SPLIT, TF.DATA PIPELINE AND CLASS WEIGHTS
- RESNET50 ARCHITECTURE AND STAGE-1 TRANSFER LEARNING
- STAGE-2 FINE-TUNING AND CALLBACK/CHECKPOINT CONTROL

MD. FARHAD HOSSAIN SHISHIR

TRAINING CONTROL + EVALUATION

- VALIDATION-QWK MODEL SELECTION AND TEST-TIME AUGMENTATION
- CLASSIFICATION REPORT, ORDINAL METRICS AND BOOTSTRAP CIS
- CONFUSION-MATRIX ERROR ANALYSIS AND GRAD-CAM EXPLAINABILITY
- FINAL RESULT INTERPRETATION, OUTPUT VERIFICATION AND PACKAGING

JOINT: PROBLEM FRAMING • FINAL REVIEW • CONCLUSION • LATEX

PROBLEM & PROJECT SCOPE

PRETOM

The final notebook predicts four retained diabetic retinopathy severity classes.

OBJECTIVE

- Classify retinal fundus images into clinically ordered DR severity levels.
- Evaluate both class-wise discrimination and ordinal agreement
- Use Grad-CAM to inspect the image regions driving predictions.

APTOS 2019 • 3,469 retained images • 224 × 224

FINAL LABEL MAPPING

0 No DR | 1 Mild | 2 Moderate | 3 PDR

COHORT CHANGE

3,662
original images

3,469
retained images

Why No Severe Class?

- Classify retinal fundus images into clinically ordered DR severity levels
- Severe class lacks proper information and the weakest link out of all categories

END-TO-END WORKFLOW

PRETOM

A REPRODUCIBLE PIPELINE FROM DATA PREPARATION TO EXPLAINABLE EVALUATION.

1

LOAD & VERIFY

APTOS METADATA + IMAGE PATHS

2

REMOVE SEVERE CLASS

REMAP PDR 4 → 3

3

EDA + QA

COUNTS, DUPLICATES, DIMENSIONS

4

PREPROCESS

RESNET50 INPUT + TRAIN-ONLY AUGMENTATION

5

SPLIT

80/10/10 STRATIFIED

6

TRAIN

TRANSFER LEARNING + FINE-TUNING

7

SELECT

HIGHEST VALIDATION QWK

8

EVALUATE

TTA + ROBUST TEST METRICS

9

EXPLAIN

CONFUSION MATRIX + GRAD-CAM

CORE PRINCIPLE: TUNE ON VALIDATION DATA, LOCK THE BEST CHECKPOINT, AND EVALUATE THE HELD-OUT TEST SET ONLY AT THE END.

DATASET, QUALITY CHECKS & PREPROCESSING

PRETOM

THE FINAL COHORT IS IMBALANCED, SO QUALITY CONTROL AND CAREFUL PREPROCESSING ARE ESSENTIAL.

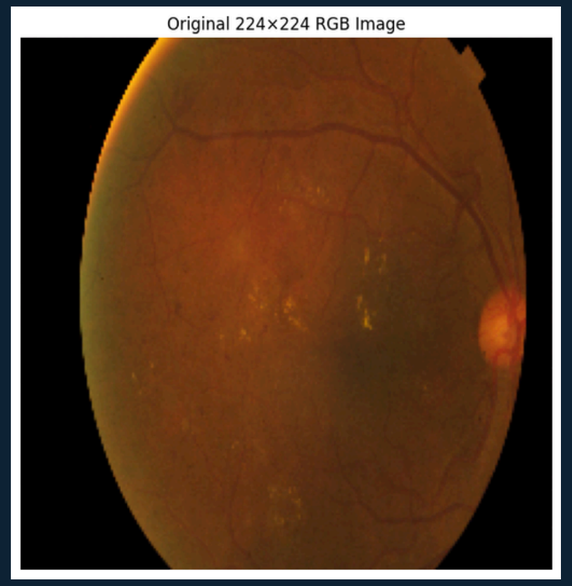
FINAL COHORT

IMAGES
3,469
after Severe removal

IMAGE SIZE
224×224
RGB

No DR	1,805	52.03%
Mild	370	10.67%
Moderate	999	28.80%
PDR	295	8.50%

PIPELINE



QUALITY CHECKS

- 0 MISSING LABELS / PATHS
- 0 DUPLICATE IDS / PATHS
- ALL RETAINED IMAGES VERIFIED AT 224 × 224

TRAINING AUGMENTATION

- Horizontal flip
- Rotation ±4%
- Zoom ±8%
- Contrast ±10%

SPLIT, INPUT PIPELINE & IMBALANCE HANDLING

PRETOM

STRATIFICATION PRESERVES CLASS PROPORTIONS; WEIGHTING REDUCES MAJORITY-CLASS DOMINANCE.

TRAIN
2,775
80% of cohort

VALIDATION
347
10% of cohort

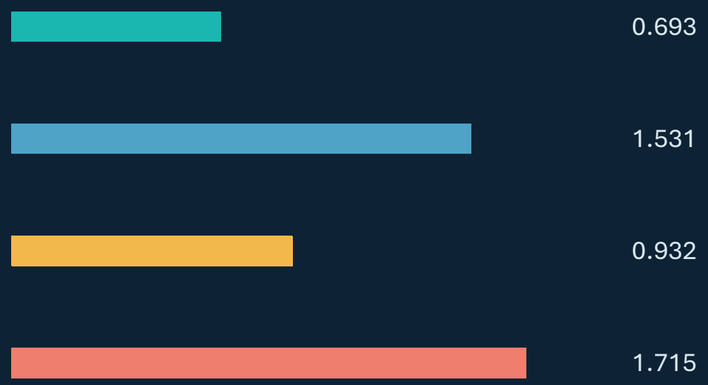
TEST
347
10% of cohort

Stratified split

random_state = 42

80 / 10 / 10

SOFTENED CLASS WEIGHTS



Training weight = √(balanced class weight)

TF.DATA PIPELINE

- DECODE RGB IMAGES AND PRESERVE 224 × 224 SHAPE
- AUGMENT TRAINING BATCHES ONLY
- APPLY RESNET50 PREPROCESS_INPUT
- BATCH SIZE = 32 + AUTOTUNE PREFETCH
- VALIDATION/TEST REMAIN DETERMINISTIC

RESNET-50 ARCHITECTURE & TRAINING STRATEGY

PRETOM

IMAGENET TRANSFER LEARNING WITH VALIDATION-QWK CHECKPOINT SELECTION.

INPUT	RESNET-50	GLOBAL AVG POOL	BATCHNORM	DENSE 256	DROPOUT	SOFTMAX
224 × 224 × 3	IMAGENET BACKBONE	2048 FEATURES	STABILIZE HEAD	RELU + L2	0.45	4 CLASSES

TOTAL PARAMETERS

24.12 M

RESNET50 + CLASSIFIER

STAGE-1 TRAINABLE

529,668

BACKBONE FROZEN

BEST VAL QWK

0.8724

CHECKPOINT METRIC

BEST VAL ACC.

0.8040

OBSERVED

STAGE 1

- Frozen ResNet50 backbone
- AdamW, LR = 3×10^{-4}, weight decay = 10^{-4}
- Up to 20 epochs; early stopped at epoch 15

STAGE 2

- Unfreeze upper 60 ResNet50 layers; keep BatchNorm frozen
- Up to 35 fine-tuning epochs with the same checkpoint rule

EVALUATION PROTOCOL

FARHAD

HELD-OUT TEST EVALUATION COMBINES TTA, CLASS-BALANCED METRICS, ORDINAL METRICS AND UNCERTAINTY.

TEST-TIME AUGMENTATION

ORIGINAL PREDICTION + HORIZONTAL-FLIP PREDICTION

AVERAGE PROBABILITIES

RAW ACC.	TTA ACC.
78.67%	81.84%
SINGLE VIEW	FINAL INFERENCE

TTA IMPROVES ACCURACY BY 3.17 PERCENTAGE POINTS.

FINAL METRICS

ACCURACY	OVERALL CORRECTNESS
BALANCED ACCURACY	MEAN CLASS RECALL
MACRO-F1	EQUAL CLASS WEIGHTING
WEIGHTED-F1	SUPPORT WEIGHTED
QWK	ORDINAL AGREEMENT
ORDINAL MAE	GRADE DISTANCE
WITHIN-ONE	±1 GRADE ACCURACY
BOOTSTRAP CI	1,000 RESAMPLES

95% BOOTSTRAP CONFIDENCE INTERVALS WERE REPORTED FOR TEST ACCURACY AND QWK.

FINAL TEST PERFORMANCE

FARHAD

THE 4-CLASS MODEL IS STRONG OVERALL, WITH THE MAIN ERRORS CONCENTRATED IN MINORITY DISEASE CLASSES.

TTA ACCURACY
81.84%
95% CI: 77.80–85.59%

BALANCED ACC.
68.97%
MEAN CLASS RECALL

MACRO-F1
0.6883
EQUAL CLASS WEIGHT

QWK
0.8975
95% CI: 0.8684–0.9247

WITHIN-ONE
98.27%
±1 GRADE

CLASS-WISE REPORT

CLASS	PRECISION	RECALL	F1	SUPPORT
NO DR	0.95	0.98	0.97	181
MILD	0.49	0.51	0.50	37
MODERATE	0.78	0.71	0.74	100
PROLIFERATIVE DR	0.53	0.55	0.54	29

MACRO AVG: PRECISION 0.69 • RECALL 0.69 • F1 0.69

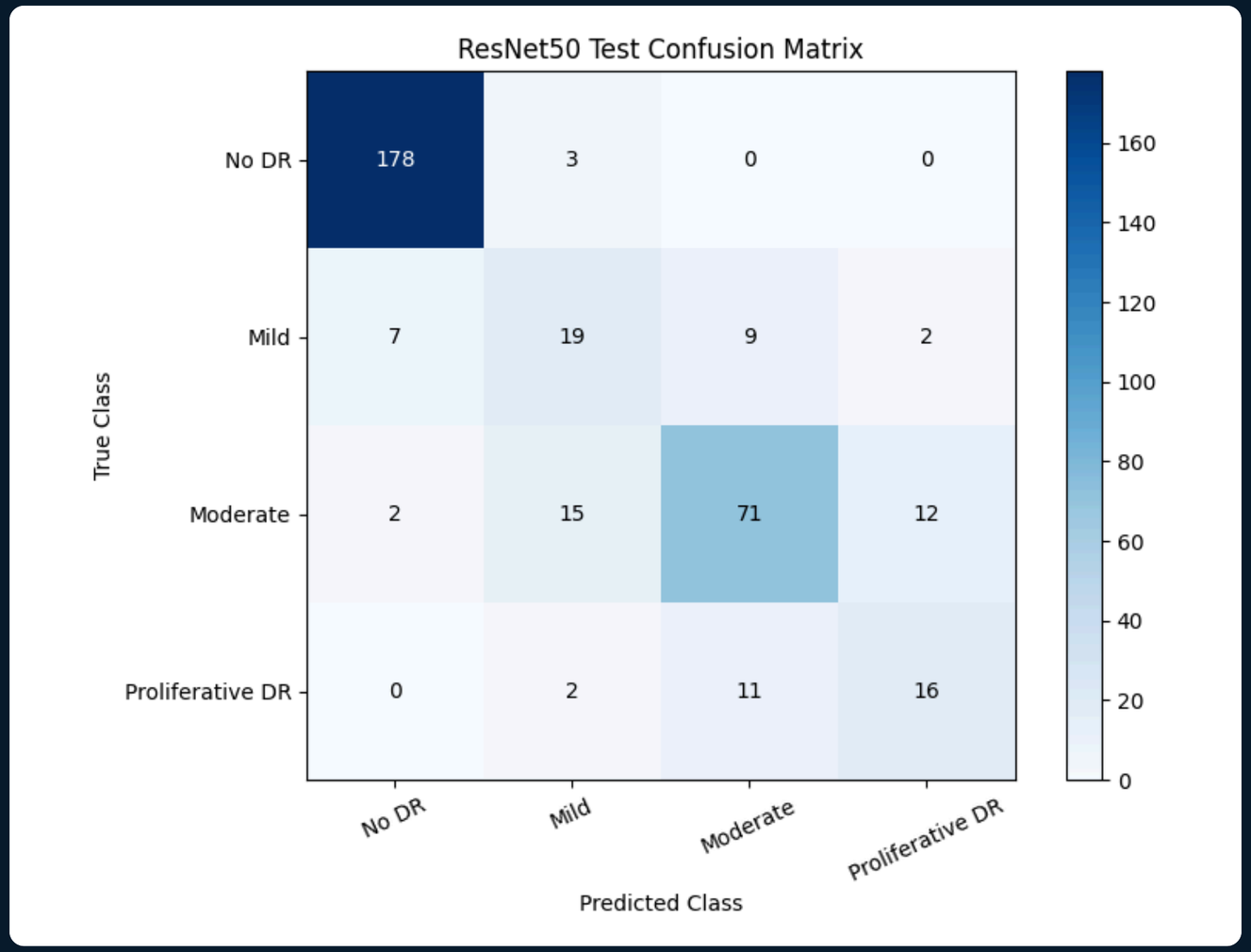
INTERPRETATION

- NO DR IS HIGHLY RELIABLE (F1 = 0.97).
- MODERATE REMAINS COMPARATIVELY STRONG (F1 = 0.74).
- MILD AND PDR REMAIN DIFFICULT BECAUSE THEY ARE MINORITY CLASSES AND VISUALLY OVERLAP ADJACENT GRADES.
- MACRO-F1 IS LOWER THAN WEIGHTED-F1, REFLECTING CLASS IMBALANCE.

ERROR ANALYSIS: CONFUSION MATRIX

FARHAD

MOST MISTAKES OCCUR BETWEEN NEIGHBORING GRADES RATHER THAN ACROSS DISTANT SEVERITY LEVELS.



KEY PATTERNS

NO DR **178 / 181 CORRECT**

98.3% RECALL

MILD **19 / 37 CORRECT**

MAIN ERRORS → MODERATE

MODERATE **71 / 100 CORRECT**

15 → MILD, 12 → PDR

PDR **16 / 29 CORRECT**

11 → MODERATE

INTERPRETATION ERRORS ARE PREDOMINANTLY LOCAL IN THE ORDINAL SCALE, WHICH IS CONSISTENT WITH THE HIGH QWK AND 98.27% WITHIN-ONE ACCURACY.

TRAINING DYNAMICS & CHECKPOINT BEHAVIOR

FARHAD

STAGE 1 PRODUCED THE BEST GENERALIZING CHECKPOINT; STAGE 2 DID NOT IMPROVE VALIDATION QWK.



BEST VAL QWK

0.8724

STAGE 1

BEST VAL ACC.

0.8040

OBSERVED

BEST VAL MACRO-F1

0.6943

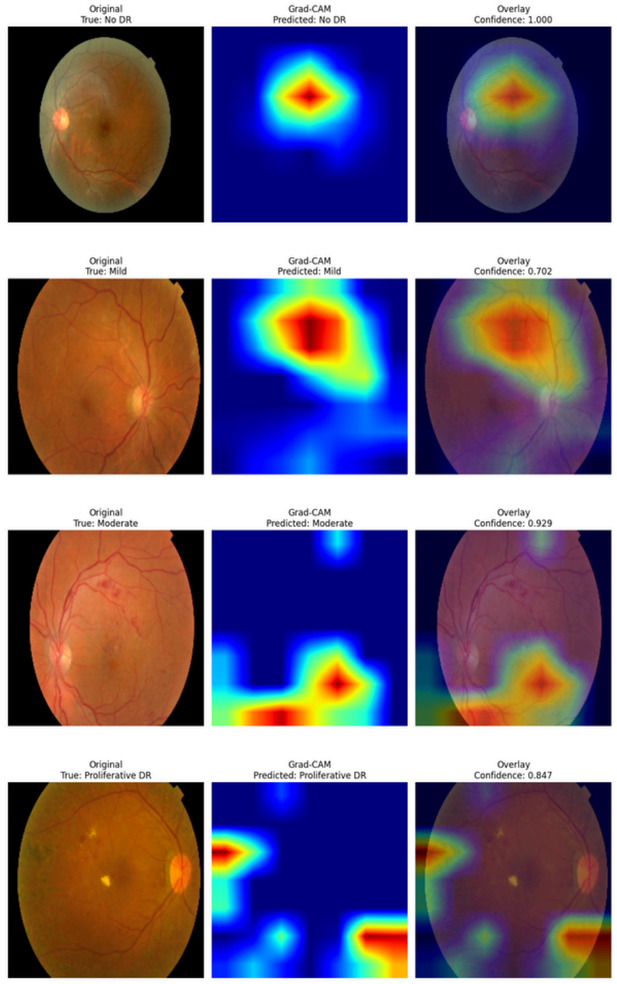
OBSERVED

CHECKPOINTING PROTECTED THE FINAL MODEL: THE LATER FINE-TUNING INSTABILITY DID NOT OVERWRITE THE BEST STAGE-1 VALIDATION CHECKPOINT.

GRAD-CAM EXPLAINABILITY

FARHAD

CLASS ACTIVATION MAPS SHOW WHERE THE FINAL RESNET50 FOCUSES WHEN MAKING PREDICTIONS.



IMPLEMENTATION

- TARGET LAYER: FINAL RESNET50 CONVOLUTION (CONV5_BLOCK3_3_CONV)
- FEATURE MAP SHAPE: 7 × 7 × 2048
- ONE REPRESENTATIVE TEST IMAGE SHOWN PER RETAINED CLASS
- HEATMAPS ARE OVERLAID ON THE ORIGINAL FUNDUS IMAGE

REPRESENTATIVE PREDICTIONS

No DR 1.000

Mild 0.702

Moderate 0.929

PDR 0.847

GRAD-CAM SUPPORTS INTERPRETABILITY, BUT IT IS QUALITATIVE RATHER THAN A CLINICAL LOCALIZATION GROUND TRUTH.

CONCLUSION & KEY TAKEAWAYS

JOINT

A CONCISE SUMMARY OF THE FINAL FOUR-CLASS EXPERIMENT.

1

REPRODUCIBLE PIPELINE

3,469 RETAINED IMAGES WERE QUALITY-CHECKED, STRATIFIED, PREPROCESSED AND TRAINED WITH FIXED SEEDS.

2

STRONG OVERALL RESULT

TTA ACCURACY REACHED 81.84% WITH QWK 0.8975 AND 98.27% WITHIN-ONE ACCURACY.

3

HONEST ERROR PROFILE

NO DR IS HIGHLY RELIABLE; MILD AND PDR REMAIN THE MAIN CHALLENGES BECAUSE OF CLASS IMBALANCE AND ADJACENT-GRADE OVERLAP.

Final message **THE MODEL DEMONSTRATES A ROBUST FOUR-CLASS DR CLASSIFICATION PIPELINE WITH STRONG ORDINAL AGREEMENT, TRANSPARENT LIMITATIONS AND EXPLAINABLE PREDICTIONS.**

THANK YOU